

SURESH KUMAAR JAYARAMAN

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EDUCATION

- Ph.D. in Mechanical Engineering** May 2018 - Aug 2021
University of Michigan, Ann Arbor, MI
Advisor: Prof. Dawn Tilbury
Thesis: Human Behavior Modeling and Human Behavior-aware Control of Automated Vehicles for Trustworthy Navigation
- Master of Science in Mechanical Engineering** Sep 2016 - Apr 2018
University of Michigan, Ann Arbor, MI
Specialization: Controls
- Bachelor of Engineering in Production Engineering** Jul 2008 - Apr 2013
Anna University, Chennai, India
Graduate with honors

RESEARCH INTERESTS

human behavior modeling, explainable decision-making, group human-robot interaction, human-robot communication, human-robot trust and teaming, interaction-aware robot control

PROFESSIONAL EXPERIENCE

- Postdoctoral Researcher at Carnegie Mellon University** May 2024 - Present
PIs: Aaron Steinfeld, Nikolas Martelaro
Topic: In-car behavior detection
- Postdoctoral Researcher at Carnegie Mellon University** Dec 2022 - Present
PIs: Aaron Steinfeld, Henny Admoni, and Reid Simmons
Topic: Explainable robot decision-making in human-robot groups
- Postdoctoral Researcher at Cornell University** Aug 2021 - Sep 2022
PIs: Guy Hoffman
Topic: Trust in human-AI teams
- Graduate Research Assistant at University of Michigan** Sep 2018 - Aug 2021
PIs: Dawn Tilbury, Lionel Robert, and Jessie Yang
Topic: Pedestrian behavior and pedestrian-AV interaction modeling
- Graduate Research Assistant at University of Michigan** Sep 2020 - Dec 2020
PI: Bogdan Epureanu
Topic: Virtual Prototyping by ADAS Tools Connected with Unreal
- Graduate Research Assistant at University of Michigan** May 2019 - Dec 2019
PIs: Dawn Tilbury, and Lionel Robert
Topic: Driver-AV interaction modeling
- Graduate Research Assistant at University of Michigan** Jan 2017 - Apr 2018
PI: Lionel Robert and Dawn Tilbury
Topic: Trust in pedestrian – automated vehicle interactions
- Graduate Researcher at University of Michigan** Jan 2017 - Dec 2017
PI: Dawn Tilbury and Kira Barton
Topic: In-process real-time monitoring of product manufacturing quality

Deputy Manager at Ashok Leyland Limited, Chennai, India <i>Prototyping and testing newly developed trucks and buses.</i>	Dec 2013 - Jul 2016
Research Fellow at Indian Institute of Technology, Madras <i>PI: Saravana Kumar Gurunathan</i> <i>Topic: Mathematical modeling of surface roughness using fractals</i>	Jul 2013 - Oct 2013
Researcher at Anna University <i>PI: Madhu Krishnan</i> <i>Topic: EEG and EMG based interfaces for design applications</i>	Jul 2012 - Apr 2013
Visiting Researcher at Indian Institute of Technology, Madras <i>PI: Saravana Kumar Gurunathan</i> <i>Topic: Brain-computer interface for design applications</i>	May 2012 - Jul 2012

PUBLICATIONS

Journal Articles

[THRI'25] (In-submission) S.K. Jayaraman, A. Steinfeld, H. Admoni, and R. Simmons, "Explaining Robot Behavior to Groups: Machine Teaching for Transparent Decision-Making." *In ACM Transactions on Human-Robot Interaction.*,

[Access'21] S.K. Jayaraman, L.P. Robert, X.J. Yang, and D.M. Tilbury, "Multimodal Hybrid Pedestrian: A Hybrid Automaton Model of Urban Pedestrian Behavior for Automated Driving Applications." *In IEEE Access*, 9, 27708-27722, 2021.

[IJSR'21] H. Azevedo-Sa, S.K. Jayaraman, C. Esterwood T., X.J. Yang, L.P. Robert, and D.M. Tilbury, "Real-time estimation of drivers' trust in automated driving systems." *In International Journal of Social Robotics*, 1-17, 2021.

[RA-L'20] H. Azevedo-Sa, S.K. Jayaraman, X.J. Yang, L.P. Robert, and D.M. Tilbury, "Context-Adaptive Management of Drivers' Trust in Automated Vehicles." *In IEEE Robotics and Automation Letters*, 5(4), 6908-6915, 2020.

[Frontiers'19] S.K. Jayaraman, C. Creech, D.M. Tilbury, X.J. Yang, A.K. Pradhan, K.M. Tsui, and L.P. Robert, "Pedestrian Trust in Automated Vehicles: Role of Traffic Signal and AV Driving Behavior." *In Frontiers in Robotics and AI*, 6, 117, 2019 - **3rd most cited paper in Frontiers in Robotics and AI, HRI section.**

Conference Proceedings

[AutoUI'25] (In-submission) "Anonymized for review." *Submitted to 2025 ACM Automotive User Interfaces and Interactive Vehicular Applications (AutoUI) Conference, 2025.*

[CSCW'25] (In-submission) "Anonymized for review." *Submitted to 2025 ACM SIGCHI Conference on Computer-Supported Cooperative Work Social Computing (CSCW), 2025.*

[IROS'24] S.K. Jayaraman, A. Steinfeld, H. Admoni and R. Simmons, "Understanding Robot Minds: Leveraging Machine Teaching for Transparent Human-Robot Collaboration Across Diverse Groups." *In 2024 International Conference on Intelligent Robots and Systems.*, Abu Dhabi, UAE, 2024.

[ICRA'20] S.K. Jayaraman, D.M. Tilbury, X.J. Yang, A.K. Pradhan, and L.P. Robert, "Analysis and prediction of pedestrian crosswalk behavior during automated vehicle interactions." *In 2020 International Conference on Robotics and Automation* (pp. 6426-6432), Virtual Event, Paris, France, 2020 - **Listed on SSRN's Top Ten download list for: InfoSciRN: Robotics (Topic) in 2020.**

[ACC'20] S.K. Jayaraman, D.M. Tilbury, X.J. Yang, A.K. Pradhan, and L.P. Robert, "Efficient Behavior-aware Control of Automated Vehicles at Crosswalks using Minimal Information Pedestrian Prediction Model." *In 2020 American Control Conference*, Virtual Event, Colorado, USA, 2020, pp. 4362-4368.

[AMRE'13] G. Sajaysurya, S.K. Jayaraman, M. Madhu, and K. Prakasan, "Electromyogram as a Viable Direct Human Input for Human Machine Interfaces." *In 2013 International Conference on Advances in Mechanical and Robotics Engineering (AMRE)*, Zurich, Switzerland, 2013.

Peer Reviewed Workshops / Symposiums / Late-breaking reports

[HRI'24] S.K. Jayaraman, A. Steinfeld, R. Simmons, and H. Admoni, "Modeling human learning of demonstration-based explanations for user-centric explainable AI." *In Workshop on Explainability for Human-Robot Collaboration*

oration, HRI 2024, Colorado, USA, 2024.

[IROS'23] S.K. Jayaraman, A. Steinfeld, H. Admoni, and R. Simmons, "Adaptive group machine teaching for human group inverse reinforcement learning." *In 3rd RL-CONFORM workshop, IROS 2023*, Detroit, USA, 2023.

[HRI'23] S. Bagchi, P. Holthaus, G. Beraldo, E. Senft, D.H. García, , Z. Han, ... and P. Pridham, "Towards Improved Replicability of Human Studies in Human-Robot Interaction: Recommendations for Formalized Reporting." *In Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, Stockholm, Sweden, 2023 - **Best LBR Nominee, 1 in 7 of 137 submissions.**

[ISTDM'21] S.K. Jayaraman, X.J. Yang, L.P. Robert, and D.M. Tilbury, "Automated Vehicle Behavior Design for Pedestrian Interactions at Unsignalized Crosswalks." *In 2021 International Symposium on Transportation Data and Modelling (ISTDM)*, Ann Arbor, MI, USA, 2021.

[HRI'20] H. Azevedo-Sa, **S.K. Jayaraman**, C.T. Esterwood, X.J. Yang, L.P. Robert, and D.M. Tilbury, "Comparing the Effects of False Alarms and Misses on Humans' Trust in (Semi) Autonomous Vehicles." *In Companion of the 2020 ACM/IEEE International Conference on Human-Robot Interaction* (pp. 113-115), Virtual Event, London, UK, 2020.

[IAVSD'19] S.K. Jayaraman, X.J. Yang, L.P. Robert, and D.M. Tilbury, "Hybrid framework for pedestrian tracking for automated vehicle applications." *In 3rd IAVSD Workshop on Dynamics of Road Vehicles: Connected and Automated Vehicles*, Ann Arbor, MI, USA, 2019.

[CPHS'18] S.K. Jayaraman, C. Creech, D.M. Tilbury, X.J. Yang, A.K. Pradhan, K.M. Tsui, and L.P. Robert, "Workload in pedestrians interacting with automated vehicles." *In 2nd IFAC Conference on Cyber-Physical and Human Systems*, Miami, FL, USA, 2018.

[HRI'18] S.K. Jayaraman, C. Creech, L.P. Robert, D.M. Tilbury, X.J. Yang, A.K. Pradhan, and K.M. Tsui, "Trust in AV: An uncertainty reduction model of AV-pedestrian interactions." *In Companion of the 2018 ACM/IEEE International Conference on Human-Robot Interaction*, Chicago, IL, USA, 2018, pp. 133-134 - **Best LBR paper, 3rd in 122 submissions.**

[RSS'17] C. Creech*, **S.K. Jayaraman***, L.P. Robert, D.M. Tilbury, X.J. Yang, A.K. Pradhan, and K. Tsui, "Trust and Control in Autonomous Vehicle Interactions." *In workshop on Morality and Social Trust in Autonomous Robots, RSS 2017*, Boston, MA, USA, 2017.

CERTIFICATIONS, HONORS, GRANTS, & AWARDS

Contributed to writing and securing a Science of Autonomy grant	2025
CMU Eberly Center Future Faculty Program	2024
Cornell Postdoc Leadership Development Program	2022
DAAD Alnet fellow in AI and Robotics – one among 41 selected worldwide	2022
Rackham Graduate School Travel Grant Award, University of Michigan	2020
College of Engineering Community Grant	2020
Cyber-Physical Human Systems Student Travel Grant	2018
Rackham Graduate School Travel Grant Award, University of Michigan	2017
Best Outgoing Student, Production Engineering, PSG College of Technology	2013

INVITED TALKS

Enhancing transparency in human-robot teams: A machine teaching approach to communicate robot decision-making to diverse human teammates

Microsoft Leaders in Robotics and AI Seminar, University of Maryland Apr 2024

Human-robot teaming: transparency and explainability

Military Institute of Engineering, Rio de Janeiro, Brazil Sep 2023

Artificial Social Intelligence for Successful Teams

Military Institute of Engineering, Rio de Janeiro, Brazil Aug 2022

Safe and Trusting Human-Autonomy Interaction

CAIRO Lab, CU Boulder Dec 2022

Multimodal Hybrid Pedestrian: A Hybrid Automaton Model of Urban Pedestrian Behavior for Automated Driving Applications

Future of Privacy Forum

Dec 2021

How can pedestrians trust autonomous vehicles (jointly with Prof. Lionel Robert)

Car 2 Car Communication Consortium

Jun 2020

A Hybrid Automaton Model of Pedestrian Behavior

U-M Mechanical Engineering Graduate Council Seminar

Jun 2020

TEACHING AND MENTORING

Guest Lectures

Decision Trees

07-180: Concepts of Artificial Intelligence, Prof. Reid Simmons, CMU

Spring 2025

Intent, Theory of Mind, and Implicit Communication in HRI

INFSCI 2300: Human Information Processing, Prof. Na Du, Univ of Pittsburgh

Spring 2024

16-867: Human-Robot Interaction, Prof. Henny Admoni, CMU

Fall 2023

Theory of Mind for Human-Robot Interaction

INFSCI 2300: Human Information Processing, Prof. Na Du, Univ of Pittsburgh

Fall 2023

INFSCI 2300: Human Information Processing, Prof. Na Du, Univ of Pittsburgh

Spring 2023

Basics of Machine Learning

15-482: Autonomous Agents, Prof. Reid Simmons, CMU

Fall 2023

CNC Machines

MAE 2250: Mechanical Synthesis, Prof. Guy Hoffman, Cornell University

Spring 2022

Design for X

MAE 2250: Mechanical Synthesis, Prof. Guy Hoffman, Cornell University

Spring 2022

Teaching Assistant

ME 461: Automatic Control. Instructor: Prof. Kira Barton, University of Michigan

Winter 2020

Held lab sessions, and office hours, prepared tutorials and assignments and helped prepare exams.

Helped in the migration to remote teaching during COVID-19.

Research Mentoring

Huy Quyen Ngo	Ph.D., Robotics, CMU		Fall'24 - Present
Aashka Pallath	B.S., A.I., CMU		Fall'24 - Present
Rithika Narayanan	B.S., CSc, Robotics, CMU		Fall'24 - Present
Shenai Chan	B.S., ECE, CMU		Sum'23 - Spr'24
JiHyun Jeong	Ph.D., Information Science, Cornell	Now Postdoc, Cornell	Fall'21 - Sum'22
Allison Duan	B.S., CSc, Cornell University	Now at Applied Intuition	Spr'22
Nam Gyu Kil	M.S., Robotics, University of Michigan		Fall'20
Eric Wiener	B.S., C.Sc., University of Michigan	Now at DoorDash	Fall'20
Kevin Fietsam	B.S., C.Sc., University of Michigan	Now at Facebook	Sum'20
Kevin Mallires	B.S., C.Sc., University of Michigan	Now at Echelon	Spr'20 - Sum'20
Alex Raistrick	B.S., C.Sc., University of Michigan	Now Ph.D. at Princeton	Sum'18

Professional Mentoring

Apoorva Roy	M.S., ME., University of Michigan	U-M GRIN Mentoring	Fall 2020
Yiting Zhang	M.S., ME., University of Michigan	U-M GRIN Mentoring	Fall 2020
Sean Vincent Herrera	Ph.D., ME., University of Michigan	UM-ME Mentoring	Fall 2019
Zhondong Liu	M.S., EECS., University of Michigan	U-M GRIN Mentoring	Fall 2018
Saylee Chandavarkar,	M.S., EECS, University of Michigan	U-M GRIN Mentoring	Fall 2018

ACADEMIC SERVICE

Organization and Leadership Activities

Associate Editor, IEEE Robotics and Automation Letters (RA-L)

Mar'24 - Present

Primary organizer of AAAI Fall symposium on "Agent teaming in mixed-motive situations"

Oct 2023

Program Committee, Technological Advances in HRI (TAHRI) Symposium

2024

Reviewer Activities

Reviewer for Conferences:

AAAI Conference on Artificial Intelligence
 ACM Conference on Human Factors in Computing System (CHI)
 ACM/IEEE International Conference on Human-Robot Interaction (HRI)
 American Control Conference (ACC)
 Conference on Robot Learning (CoRL)
 IEEE International Conference on Intelligent Robots and Systems (IROS)
 IEEE International Conference on Robotics and Automation (ICRA)
 IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)
 IFAC Conference on Cyber-Physical and Human-Systems (CPHS)
 Robotics: Science and Systems (RSS) Pioneers

Reviewer for Journals:

ACM Social Computing
 ACM Transactions on Human-Robot Interaction (THRI)
 Behaviour and Information Technology (BIT)
 Digital Threats: Research and Practice (DTRAP)
 Ergonomics
 Frontiers in Psychology
 IEEE Robotics and Automation Letters (RA-L)
 IEEE Transactions on Automation Science and Engineering (TASE)
 IEEE Transactions on Control Systems Technology (TCST)
 IEEE Transactions on Intelligent Transportation Systems (T-ITS)
 IEEE Transactions on Systems, Man, and Cybernetics (TSMC)
 Journal of Cognitive Engineering and Decision Making (JCEDM)
 Journal of Public Transportation (JPT)
 MDPI Artificial Intelligence
 MDPI Ergonomics
 MDPI Information
 MDPI Robotics
 Simulation Modelling Practice and Theory (SIMPAT)
 Transportation Research Part C (TRC)

Other service Activities

Member of HRI METRICS Working Group, IEEE P3108	Feb 2022 – Present
Member of CMU-RI Climate Committee for Diversity, Equity, and Inclusion	Jan 2023 – Present
Student Volunteer, International Conference on Human-Robot Interaction (HRI)	2021
Student Volunteer, International Conference on Robotics and Automation (ICRA)	2020
Student Volunteer, K-12 Outreach program, Detroit Area Pre-College Engineering Program (DAPCEP)	2018 - 2019

MEDIA

"Linking self-driving cars to traffic signals might help pedestrians give them the green light" The Conversation	Apr 2020
"How can pedestrians trust autonomous vehicles" University of Michigan's Robotics News	Jan 2020
"Stop, look and look out: Pedestrians' trust of autonomous vehicles at crosswalks" University of Michigan, School of Information News	Apr 2018
"Agent teaming in mixed motive situations – an AAAI symposium" AIHub	Jan 2024

REFERENCES

References available upon request.